KOR STRUCTURAL — OPERATIONS

Immutability Server — Veeam Hardened Repository Build Sheet

A local, ransomware-proof backup copy — the fast on-prem recovery tier

Date: 2026-07-27 · **Owner:** Ian Lalonde (KOR) · One-page build reference

> **Why this box.** Immutability is enforced by the Linux kernel **on this machine**, so even a compromised domain or a compromised BK01 cannot delete or encrypt these backups. It is your fast, on-prem, ransomware-proof recovery copy — separate from the off-site copy in Toronto.

1. Hardware — repurpose an old PC + new drives

Part	Spec	Notes
Chassis / PC	64-bit, **16 GB RAM** (32 GB better), room for **4 drives**	Tower/workstation, not an SFF/NUC. Pick the most reliable old box.
SATA ports	4–6 SATA ports	If short, add a **~$40 HBA card** (IT-mode, e.g. LSI 9211-8i).
PSU	Enough SATA power leads for 4 drives	Splitter if needed.
Drives	**4 × Seagate Exos X18 16 TB SATA** (or IronWolf Pro 16 TB)	**MUST be CMR + SATA.** Never SMR, never SAS. ~$280–300 ea. Optional 5th as cold spare.
NIC	Onboard gigabit is fine	(This box isn't on the 10G core anyway.)

Storage layout: 4 drives → **software RAID (mdadm) RAID5** ≈ **48 TB usable** → **XFS** (with reflink) on the array. Redundancy matters — this is your only local immutable copy.

2. Operating system & filesystem

- **OS:** Ubuntu Server **22.04 LTS** (or newer LTS). Minimal install, no GUI.
- **NOT domain-joined.** Local accounts only.
- **Filesystem: XFS** on the mdadm array, mounted with reflink (gives Veeam fast-clone — space-efficient synthetic fulls without ReFS's RAM cost).
- **Dedicated repo mount**, e.g. `/mnt/veeamrepo`.

3. Hardening — this is what makes it immutable

- Create a **single-use Veeam repository account** (Veeam manages it; you never log in with it).
- Set the **immutability period: 14–30 days** (recommend 30 for ransomware dwell-time).
- **SSH:** key-only or disabled after setup; no root login; firewall to Veeam/BK01 only.
- **Time sync (NTP)** must be correct — immutability is time-based.
- Keep it **physically in the server room**, powered via the UPS.

4. Network

- **Static IP in the reserved band** `192.168.1.200-.254` (outside the DHCP pool `.51-.199`) — e.g. `.241`. Document it in the network sheet.
- Gateway `192.168.1.1`, DNS `192.168.1.30`.
- LAN is fine for now (immutability is the protection, not network isolation); move to a management VLAN once VLANs exist.

5. Wire it into Veeam — I'll drive this part with you

1. Veeam console → **Backup Infrastructure** → **Backup Repositories** → **Add** → **Linux (Hardened Repository)** → point at `/mnt/veeamrepo`, enter the single-use creds, tick **"Make backups immutable for N days."**
2. Create a **Backup Copy job** → source = existing local backups → target = this hardened repo. This becomes your on-prem immutable tier.
3. Optional later: wrap Synology + hardened repo + object storage into a **Scale-Out Backup Repository** with GFS.

6. Buy / do checklist

- [ ] Pick the PC (16 GB+ RAM, 4 bays, best PSU/board)
- [ ] Order **4 × Exos X18 16 TB SATA** (+ optional 5th spare); HBA card if ports are short
- [ ] Install Ubuntu 22.04 LTS
- [ ] Build mdadm RAID5 + XFS on `/mnt/veeamrepo`
- [ ] Harden (single-use account, SSH lockdown, NTP, firewall)
- [ ] Assign static IP `.241`, document it
- [ ] **Then ping me** — I'll walk the Veeam Hardened Repository add + Backup Copy job

Cost: ~$1,100–1,300 (drives) + $0 (repurposed PC) + optional ~$40 HBA. That's your entire local ransomware-recovery tier.